M.TECH CSE · IIT ROORKEE · GROUP 5

Forecasting service demand for a plumbing chain

A machine learning model and planning app that predicts daily service calls per branch and service type, up to 28 days ahead

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THE PROBLEM

Why forecast demand?

Technicians are scheduled days ahead, but calls arrive with the weather.

One cold night can multiply burst-pipe calls twentyfold. Staffing for an average day leaves the busiest days uncovered.

Too few technicians

Missed emergency calls, overtime, lost revenue

Too many technicians

Paid hours with no jobs to do

Our goal

Predict daily calls per branch and service, 1–28 days ahead, with a likely range

THE DATA

A synthetic dataset modelled on a real chain

8
branches in 8 US cities

9
service types, from drains to repiping

3 yrs
of daily data, 2023–2025

74k
branch × service × day rows

Six linked tables

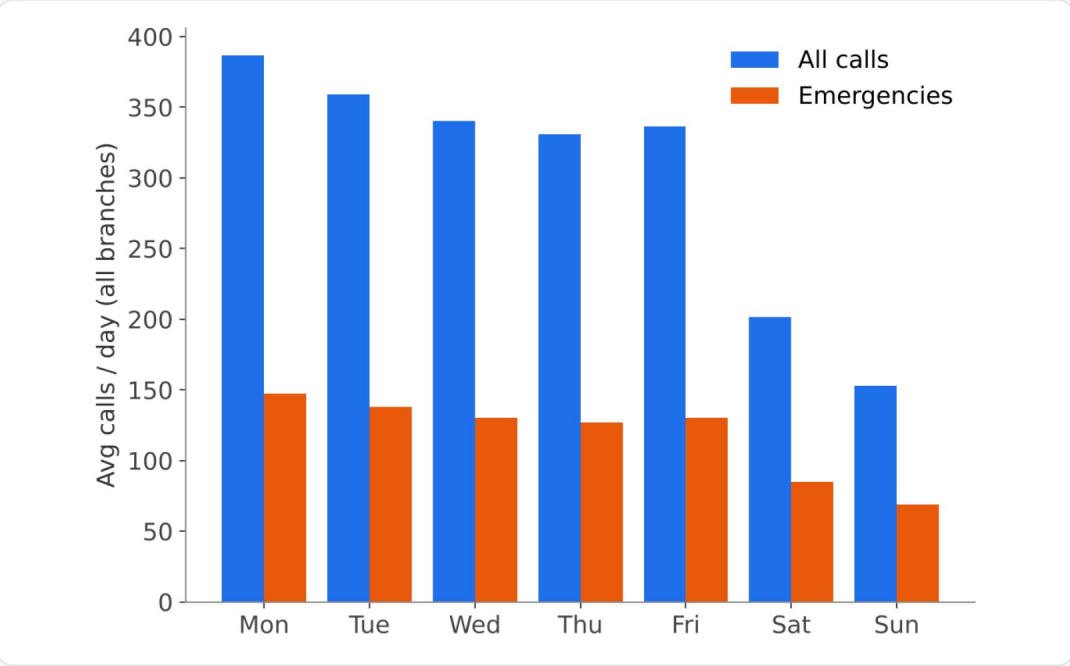
Daily demand (target: calls received), weather, marketing spend, holidays, branch details and 240k individual job records.

Realistic on purpose

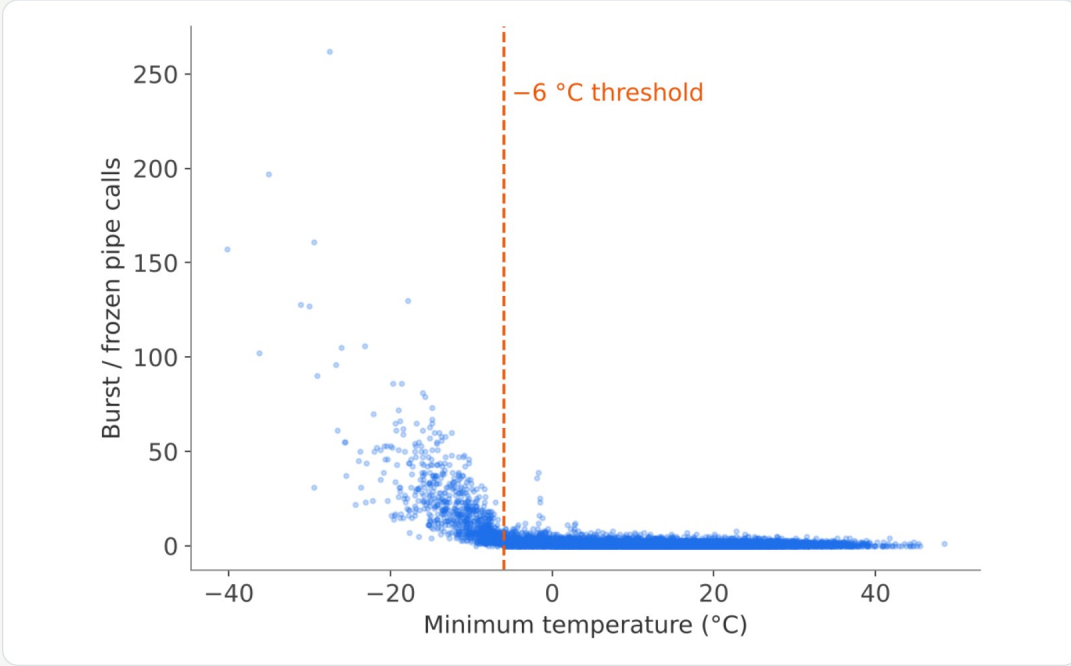
Built-in freeze spikes, holidays and a new branch, plus sensor gaps and a 3-day data outage. Published on Kaggle under CC0.

EXPLORING THE DATA

What drives demand



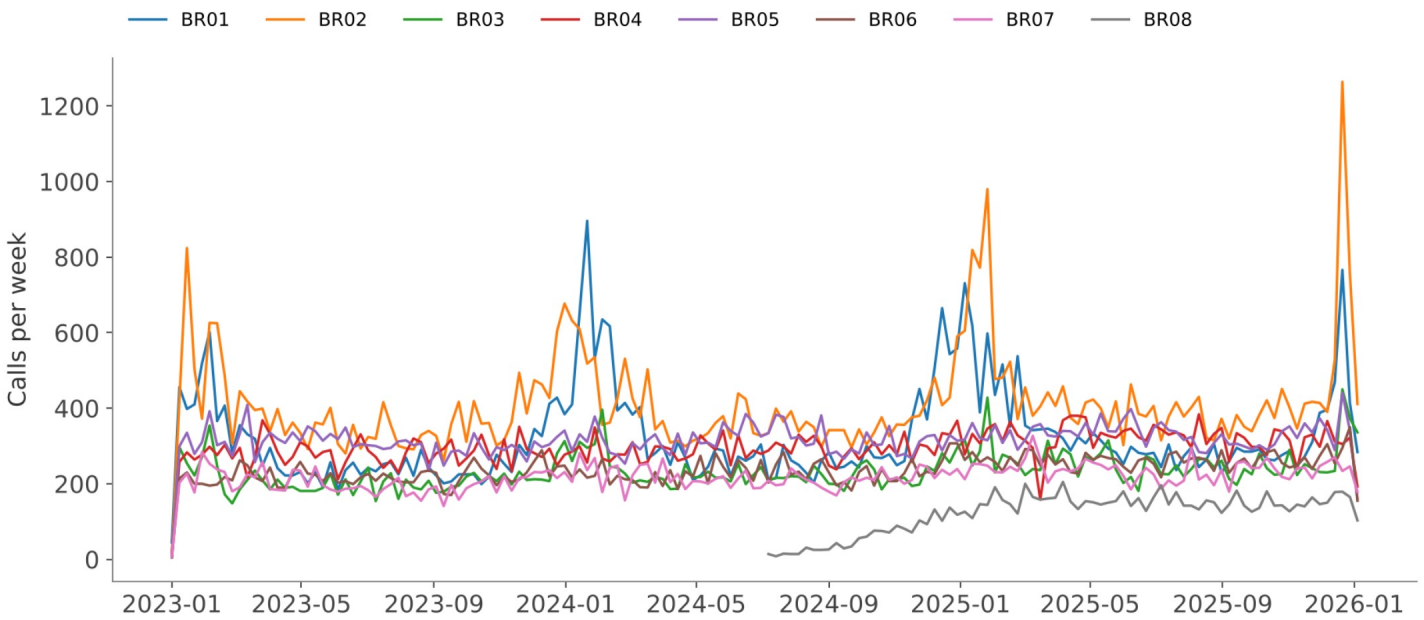
Weekday: scheduled work peaks on Monday; emergencies barely change



Weather: burst-pipe calls jump once nights fall below about −6 °C

EXPLORING THE DATA

Weekly calls per branch, 2023–2025



Growth: every branch trends upward by 3–10% a year

Winter peaks: sharp spikes in cold cities during freezes

New branch: BR08 Charlotte opens in July 2024 with no history

METHOD

From raw tables to a forecast

Join
6 tables on date and branch

→

Clean
Fill weather gaps and the outage

→

Features
34 inputs in four groups

→

LightGBM
Main model plus a burst-pipe model

→

Forecast
Expected calls and a P10–P90 range

Feature group	Examples
Calendar	Weekday, season, holidays, day after Thanksgiving, Christmas week
Weather	Min temperature, freeze severity, thaw day, 3-day rainfall
Marketing	Daily spend, carry-over effect of past spend, promotions
Demand history	Calls 4–7 weeks ago and a year ago, 28-day rolling average

METHOD

The machine learning models we used

1 · Main forecaster

LightGBM regressor

Gradient-boosted decision trees with a Poisson objective for counts. One global model for all 72 branch × service series: 800 trees, 63 leaves each.

2 · Specialist

Burst-pipe LightGBM

Same algorithm, trained only on burst and frozen pipe rows. Replaces the main model for that service, which is rare and very spiky.

3 · Likely range

Quantile LightGBM × 2

Quantile loss at α = 0.1 and 0.9 gives the low (P10) and high (P90) estimate for every forecast.

4 · For comparison

Baselines

Seasonal naive methods (same weekday 4 weeks ago, average of last 4 same weekdays, same day last year) and our first LightGBM with a Tweedie objective.

Why gradient-boosted trees?

- Learns thresholds like −6 °C
- Captures interactions such as service × weather
- Handles categories and missing values
- Trains in about 20 seconds

METHOD

Testing it honestly



The test period is never seen during training: 13,248 rows

Split by time

A random split would let the model see the future and overstate accuracy.

No leakage

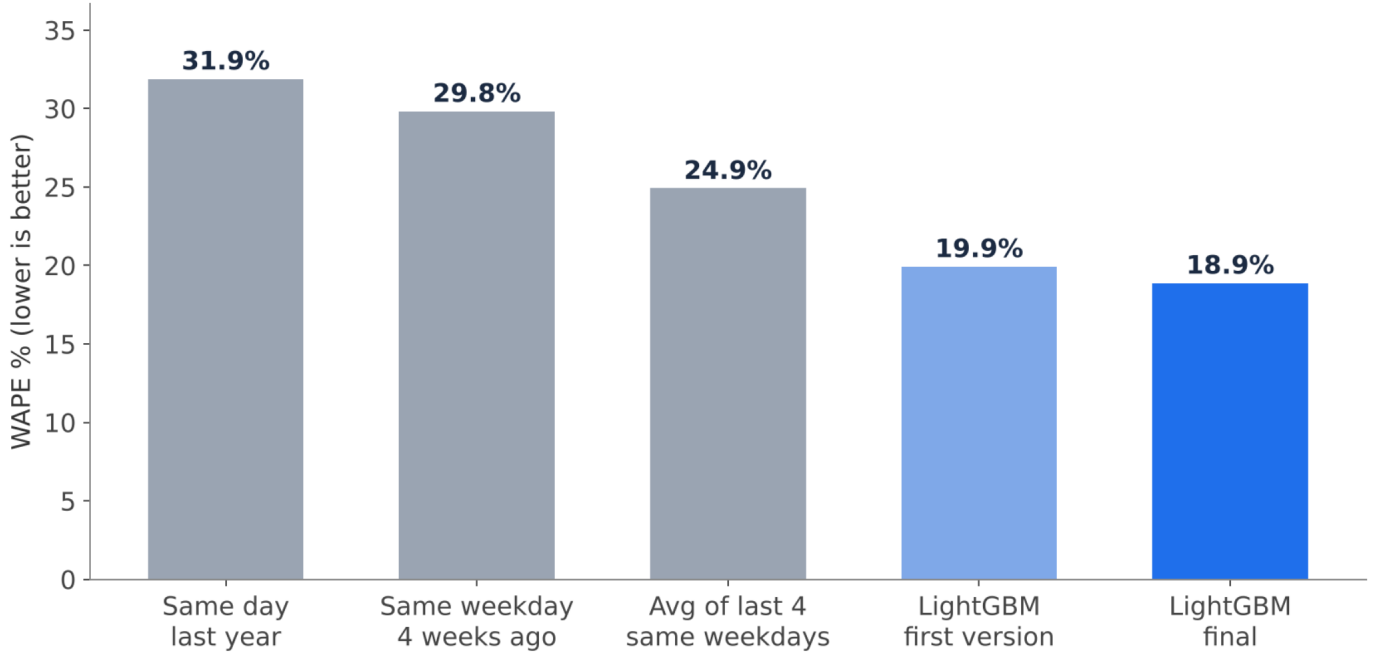
Every past-demand feature is at least 28 days old, so it is known when forecasting 4 weeks ahead.

WAPE metric

Total absolute error ÷ total actual calls. Works when many days have zero calls.

RESULTS

Accuracy on the unseen test period



18.9%

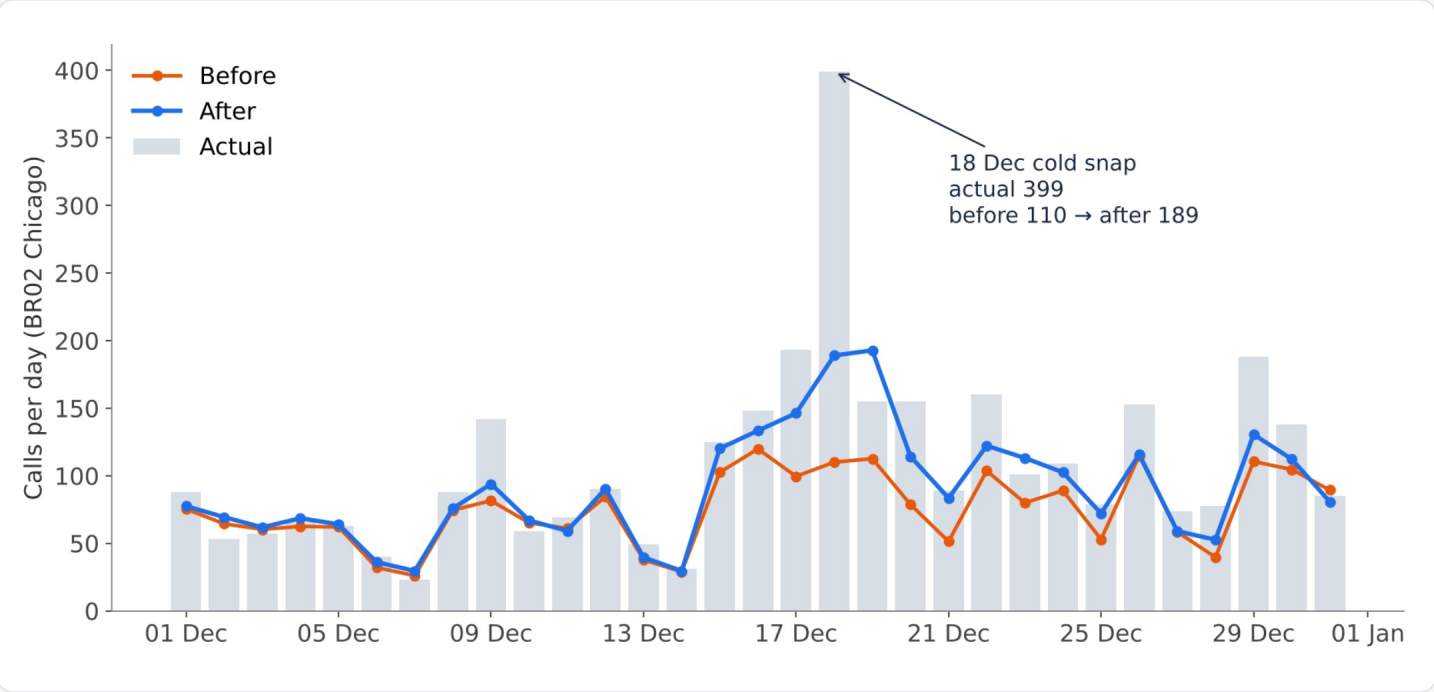
error per branch per day (WAPE)

24% less error than the best simple method (24.9%)

Per single service per day: 48.1% vs 56.2%. Small daily counts (often 0–5 calls) are mostly random.

ERROR ANALYSIS

Fixing the cold-snap miss



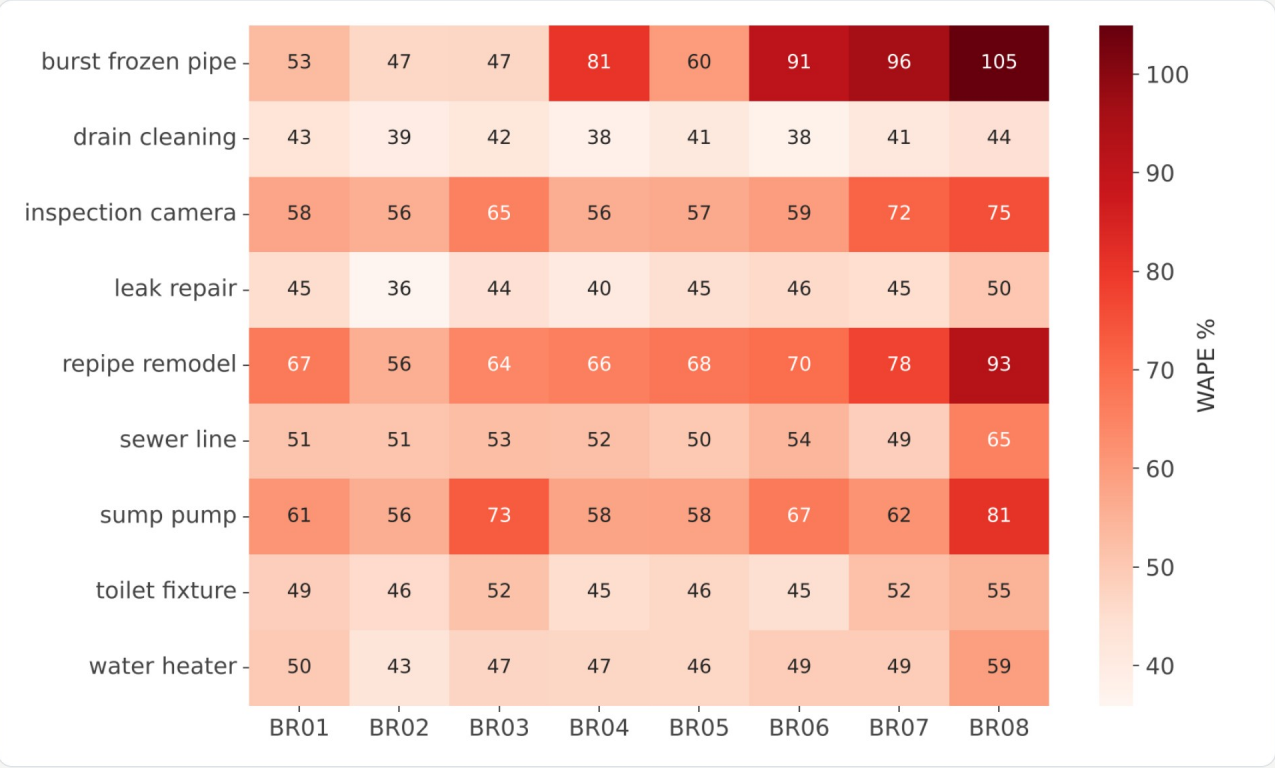
The first model was **30% low in December** at BR02 because freeze days are rare in training.

1. Freeze features only for services that react to cold
2. Poisson objective for count data
3. A separate model for burst pipes

Peak day: 110 → 189 (actual 399)

ERROR ANALYSIS

Where the remaining error sits



Most accurate

Drain cleaning and leak repair, about 40%: frequent, steady jobs

Least accurate

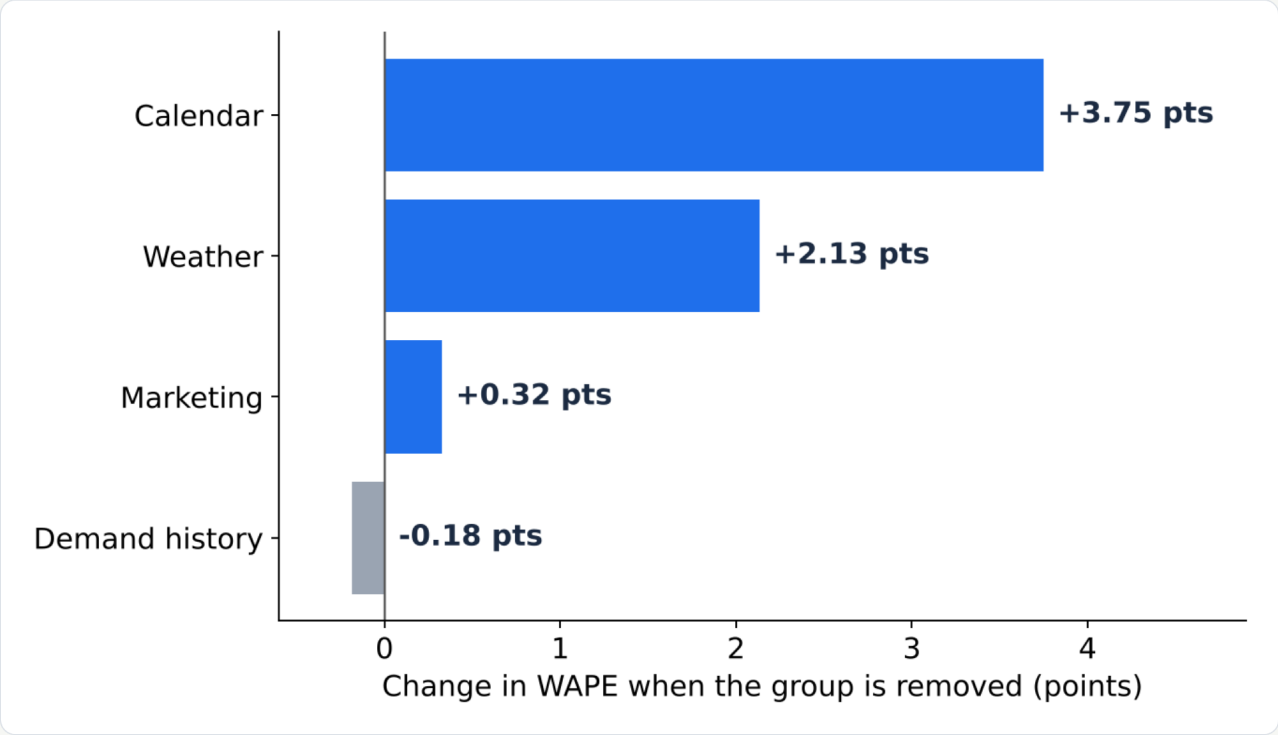
Burst pipes in mild cities (89–108%), where freezes are rare and spiky

Also weak

Remodels (rare, large jobs) and BR08, the branch with the least history

WHAT THE MODEL RELIES ON

Removing one feature group at a time



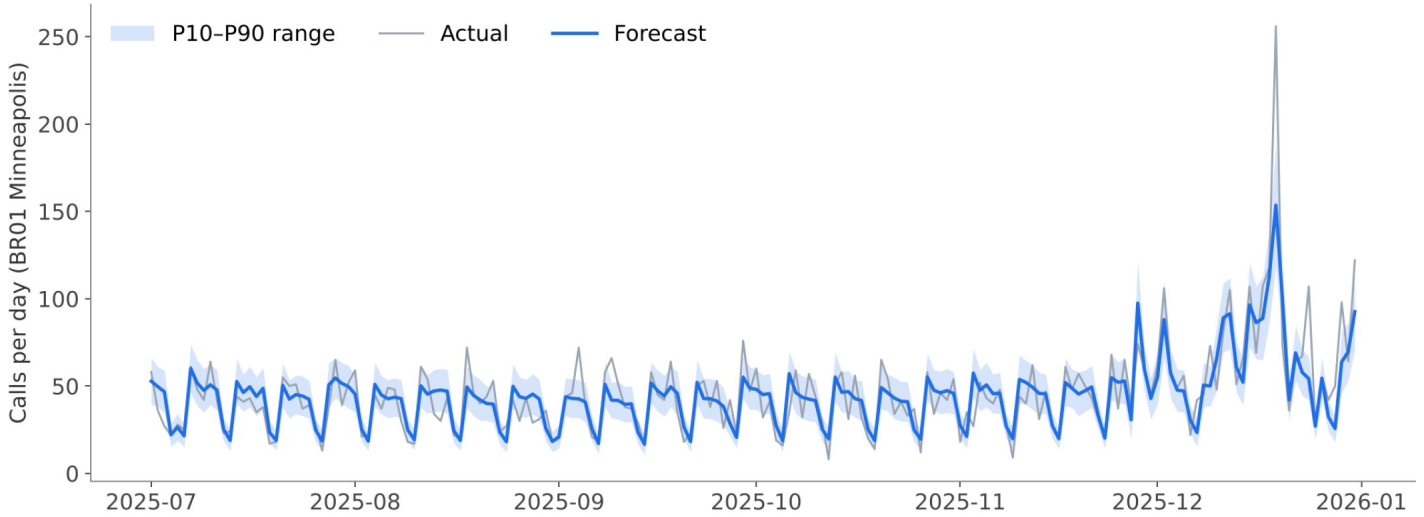
Calendar matters most, then weather. Marketing adds a small, real gain.

A limit of synthetic data

Demand history adds nothing here because our data was generated from the other drivers. With real data it usually matters more, capturing events no feature describes.

UNCERTAINTY

A likely range, not just one number



73%

of actual values fall inside the P10–P90 range (target 80%)

P90

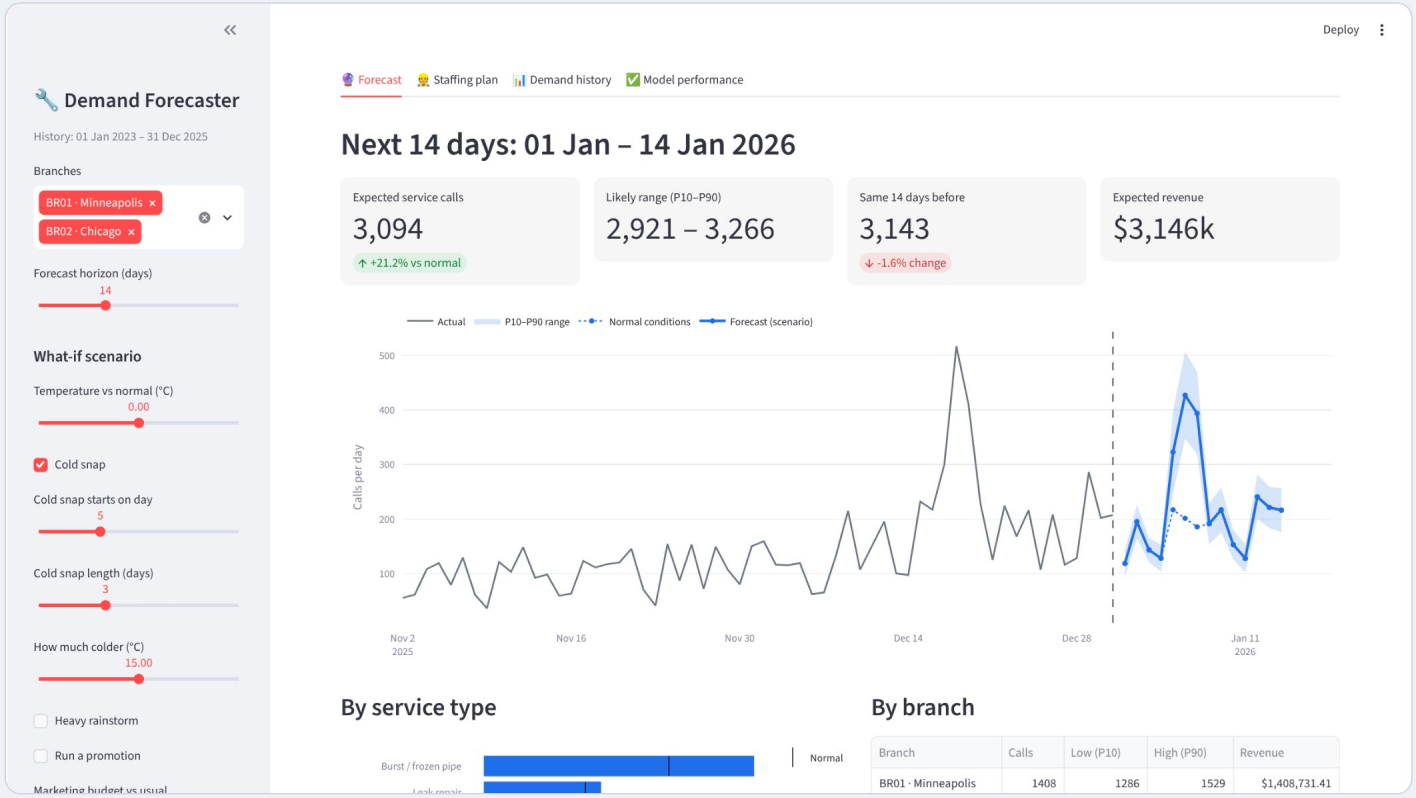
is the number to staff from on cold-snap days

$\sqrt{\Sigma\sigma^2}$

combines service ranges into branch totals, rather than adding percentiles

THE PLANNING APP

Forecasts with what-if scenarios



Built in **Streamlit** on top of the trained model.

What-if controls

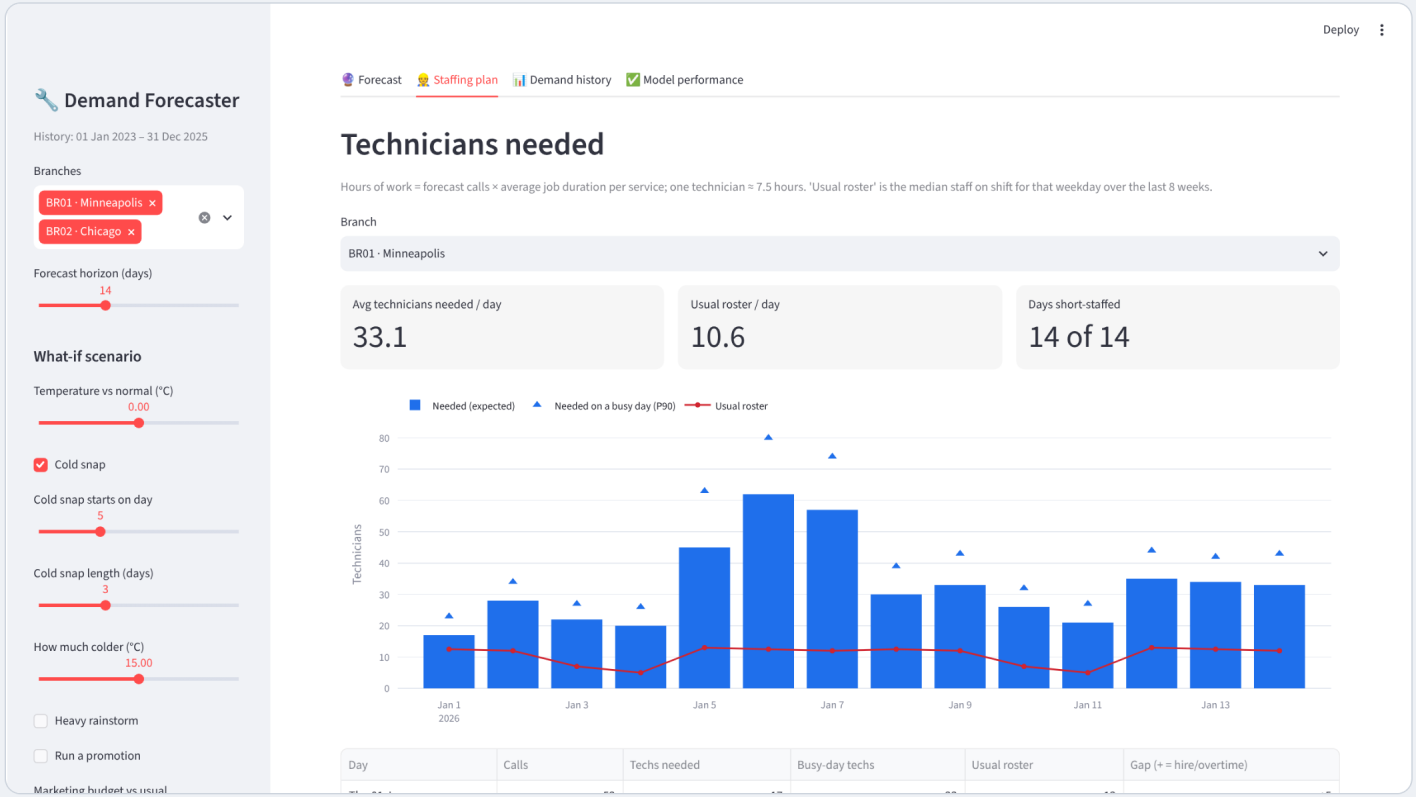
Temperature shift, cold snap, rainstorm, promotion days, marketing budget

Example

A 3-day cold snap raises 14-day calls for Minneapolis and Chicago by 21%

THE PLANNING APP

From calls to a staffing plan



forecast calls × average job hours ÷ 7.5-hour shift = **technicians needed**

Compared with the usual roster for that weekday, it shows the gap to cover with hiring or overtime.

The large winter gap reflects how the synthetic data was generated, not real staffing levels.

LOOKING AHEAD

Limitations and next steps

Limitations

- Synthetic data, so results need confirming on real company records
- Extreme cold-snap peaks are still under-predicted
- P10–P90 range is slightly too narrow (73% vs 80%)
- Future weather comes from past averages, not a real forecast

Next steps

- Tune settings with time-series cross-validation
- Compare with Prophet and SARIMAX
- Connect a live weather-forecast feed
- Feed the staffing plan into technician scheduling

Thank you

18.9% error per branch per day, 24% better than the best simple method, inside an app a manager can use.

Questions?

Dataset: kaggle.com/datasets/adityaraiiiii/plumbing-demand-dataset